

공학수학 I (BS102a-05)

학기	2026년 1학기
담당교수	Carlos Scarinci
이수구분	공통필수
학점	3.0
강의시간	화11:00-13:00(E7 - 223), 목11:00-13:00(E7 - 223)
강의실	E7 L10 (교수실)
소속	기초학부

교과개요

Calculus and Differential Equations are essential mathematical tools in science and engineering. They are used to model and understand physical phenomena, allowing us to predict the behavior, evolution, and outcomes of various processes in both natural and engineered systems. In this course, we will review the fundamentals of differential and integral calculus, develop methods for solving ordinary differential equations, and apply these techniques to analyze key models in science and engineering.

교과목표

By the end of the course, students should be able to solve first- and second-order ordinary differential equations using standard analytical methods, analyze linear and nonlinear systems qualitatively, apply Laplace transforms to initial value problems, and interpret solutions in scientific and engineering contexts.

주차별 강의계획

1주차:Calculus Review
2주차:Calculus Review
3주차:First Order ODEs
4주차:First Order ODEs
5주차:Linear ODEs
6주차:Homogeneous Linear ODEs
7주차:Homogeneous Linear ODEs
8주차:Midterm Exam Week
9주차:Nonhomogeneous Linear ODEs
10주차:Linear Algebra Basics
11주차:Linear Systems of ODEs
12주차:Linear Systems of ODEs
13주차:Nonlinear Autonomous Systems
14주차:Laplace Transforms
15주차:Laplace Transforms
16주차:Final Exam Week

수업방법

The course will consist primarily of blackboard lectures covering material from the textbooks. A list of suggested exercises will be provided each week. Students are encouraged to read the relevant material and attempt the exercises before class, and to actively engage during lectures by asking questions and participating in discussions.

학습윤리

Cheating and plagiarism are serious violations of academic integrity and will not be tolerated. Any student found responsible for cheating will automatically fail the course. This also applies to those who knowingly allow their work to be copied.

교과정책(방침)

Any student who misses 25% or more classes will receive a failing grade. Justified absences will be considered as attended. Attendance at both the midterm and final examinations is mandatory. Absence from either examination will result in a failing grade.

준비물 및 기타

Single-variable calculus, basic integration techniques, and familiarity with exponential and trigonometric functions. Students must have regular access to a laptop computer for assignments and computational demonstrations as required.